

Enterprise Pattern: Unit of Work



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Coming Up



Detailing the unit of work pattern

EF Core and the unit of work pattern

Real-world use cases



Detailing the Unit of Work Pattern

What if you want to ensure that the data changes you want to persist across different repositories are handled in the same transaction?



Unit of work pattern

The primary goal of this pattern is to manage transactions and coordinate multiple data operations within a single logical unit



**The unit of work ensures
that either all operations
within the unit succeed or
none of them do**



```
public interface IUnitOfWork
{
    Task CommitAsync();
    void Rollback();
}

public abstract class UnitOfWork(OrderDbContext orderDbContext) : IUnitOfWork
{
    // TODO
}
```

Detailing the Unit of Work pattern



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Detailing the Unit of Work pattern

Interface defines methods to commit changes in one transaction, and to roll them back



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public interface IUnitOfWork
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public abstract class UnitOfWork(OrderDbContext orderDbContext) : IUnitOfWork
{
    // TODO in the demo
}
```

Detailing the Unit of Work pattern

Interface defines methods to commit changes in one transaction, and to roll them back
Base class implementation implements the `IUnitOfWork`



```
public class CreateOrderWithOrderLinesUnitOfWork(OrderDbContext orderDbContext,
    IRepository<Order> orderRepository,
    IOrderLineRepository orderLineRepository) : UnitOfWork(orderDbContext)
{
    // TODO in the demo
}
```

Detailing the Unit of Work pattern



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public class CreateOrderWithOrderLinesUnitOfWork(OrderDbContext orderDbContext,
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    IOrderLineRepository orderLineRepository) : UnitOfWork(orderDbContext)
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    // TODO in the demo
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```

Detailing the Unit of Work pattern

Actual unit of work depends on the `DbContext` & one or more repositories



Demo



Implementing the unit of work pattern



**EF Core provides an
implementation of the unit
of work pattern**

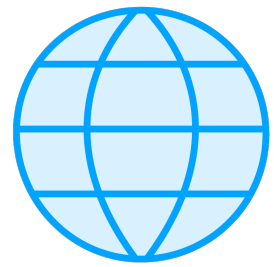


EF Core and the Unit of Work Pattern

The `DbContext` in EF Core acts as the unit of work, as it manages the transaction scope



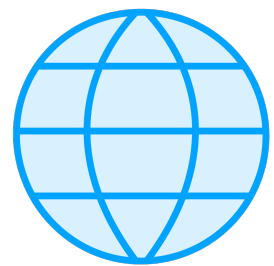
Real-world Use Cases



Banking application



E-commerce checkout process



... and almost all applications that deal with data access



Summary



The primary goal of the unit of work pattern is to manage transactions and coordinate multiple data operations within a single logical unit



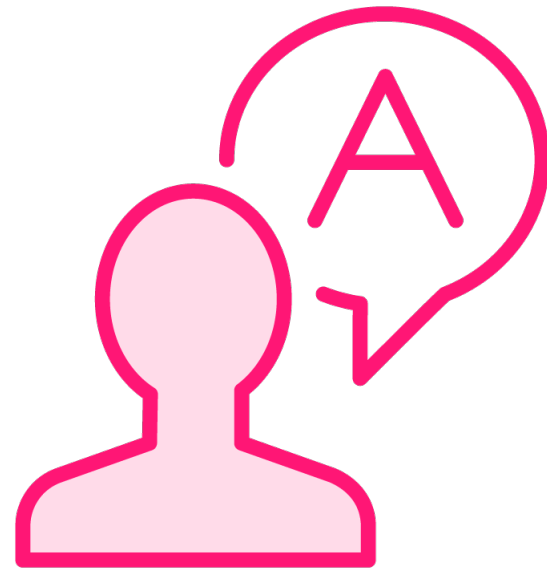
Summary



EF Core's `DbContext` is an implementation of this pattern



The End is Nigh...



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to be
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